# Pediatric Difficult Airway

## Section I: Scenario Demographics

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|---|---|
| Scenario Title: | Pediatric Difficult Airway |
| Date of Development: | 31/10/2017 (DD/MM/YYYY) |
| Target Learning Group: | ☐ Juniors (PGY 1 – 2) ☐ Seniors (PGY ≥ 3) ☒ All Groups |

## Section II: Scenario Developers



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|---|---|
| Scenario Developer(s): | Jonathan Pirie |
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## Section III: Curriculum Integration

| Learning Goals & Objectives | |
|---|---|
| Educational Goal: | Expose learners to the resuscitation of a child with difficult airway requiring needle cricothyroidotomy - a rare but critical skill in the ED. |
| CRM Objectives: | 1) Effectively and calmly lead a team through a failed airway algorithm necessitating percutaneous cricothyroidotomy<br>2) Demonstrate high quality closed-loop communication throughout resuscitation of an infant |
| Medical Objectives: | 1) Verbalize and work through the differential diagnosis of stridor<br>2) Identify the need for intubation in a stridorous patient<br>3) Prepare options for airway management if intubation fails<br>4) Demonstrate the technique for needle cricothyroidotomy |

### Case Summary: Brief Summary of Case Progression and Major Events

The ED team is called to manage a 2-year-old boy in severe respiratory distress with stridor and hypoxia. Initial management steps (humidified O2, nebulized epinephrine and dexamethasone) fail to improve the patient's respiratory status, and the team must prepare for a difficult intubation. They will encounter difficulties with both bagging and passing the endotracheal tube due to airway edema, which will necessitate an emergency needle cricothyroidotomy.

### References


Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.

Otriz-Alvarez, O. (2017). Acute management of croup in the emergency department. *Paediatr Child Health*. 22(3): 166-169.

The Airway Jedi (2017). Tips and tricks for intubation, airway management, anesthesia, and patient safety. Accessed on Nov 2, 2017 at https://airwayjedi.com/



## Section IV: Scenario Script

| A. Scenario Cast & Realism | | | |
|---|---|---|---|
| Patient: | ☒ Computerized Mannequin | Realism: | ☒ Conceptual |
| | ☐ Mannequin | | ☐ Physical |
| | ☐ Standardized Patient | *Select most important dimension(s)* | ☐ Emotional/Experiential |
| | ☐ Hybrid | | ☐ Other: |
| | ☐ Task Trainer | | ☐ N/A |

| Confederates | Brief Description of Role |
|---|---|
| Nurse | To assist at bedside, cue team to patient's status PRN |
| RT | To assist at bedside for airway management |
| | |

| B. Required Monitors | | |
|---|---|---|
| ☒ EKG Leads/Wires | ☐ Temperature Probe | ☐ Central Venous Line |
| ☒ NIBP Cuff | ☒ Defibrillator Pads | ☒ Capnography |
| ☒ Pulse Oximeter | ☐ Arterial Line | ☐ Other: |

| C. Required Equipment | | |
|---|---|---|
| ☒ Gloves | ☒ Nasal Prongs | ☐ Scalpel |
| ☒ Stethoscope | ☒ Venturi Mask | ☐ Tube Thoracostomy Kit |
| ☒ Defibrillator | ☒ Non-Rebreather Mask | ☒ Cricothyroidotomy Kit |
| ☒ IV Bags/Lines | ☒ Infant Bag Valve Mask | ☐ Thoracotomy Kit |
| ☒ IV Push Medications | ☒ Laryngoscope | ☐ Central Line Kit |
| ☐ PO Tabs | ☒ Video Assisted Laryngoscope | ☐ Arterial Line Kit |
| ☐ Blood Products | ☒ Infant ET Tubes | ☐ Other: |
| ☒ Intraosseous Set-up | ☒ LMA | ☐ Other: |

| D. Moulage |
|---|
| None |

| E. Approximate Timing | | | | | |
|---|---|---|---|---|---|
| Set-Up: | 5 min | Scenario: | 15 min | Debriefing: | 20 min |

## Section V: Patient Data and Baseline State

| A. Clinical Vignette: To Read Aloud at Beginning of Case |
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| You are working in the ED, and your team has been called urgently to see a 2-year-old old boy with difficulty breathing. The patient was brought in by his mother, who states he's had a 2-day history of runny nose. Today he developed a barking cough with fever, and is "breathing with a funny noise". |

| B. Patient Profile and History | | | | |
|---|---|---|---|---|
| Patient Name: Isaac | Age: 2 years | | Weight: 12 kg | |
| Gender: ☒ M ☐ F | | | | |
| Chief Complaint: Respiratory distress | | | | |
| History of Presenting Illness: 2 day history of runny nose. Today developed barking cough with fever and is now "breathing funny" | | | | |
| Past Medical History: | Previously healthy | Medications: | Tylenol, last dose 7 hours ago | |
| | Fully immunized | | | |
| Allergies: None | | | | |
| Social History: Lives with mother, in daycare during day. | | | | |
| Family History: Non-contributory | | | | |
| Review of Systems: | CNS: | Alert | | |
| | HEENT: | No facial/ mouth swelling | | |
| | CVS: | Nil | | |
| | RESP: | Audible stridor at rest | | |
| | GI: | Nil | | |
| | GU: | Nil | | |
| | MSK: | Nil | INT: | Nil |

| C. Baseline Simulator State and Physical Exam | | | |
|---|---|---|---|
| ☐ No Monitor Display | ☒ Monitor On, no data displayed | ☐ Monitor on Standard Display | |
| HR: 180/min | BP: 80/46 | RR: 65/min | $O_2$SAT: 90% RA |
| Rhythm: Sinus tach | T: 38.4°C rectal | Glucose: 5 mmol/L | GCS: 15 (E4 V5 M6) |
| General Status: Unwell | | | |

| | | | |
|---|---|---|---|
| CNS: | Alert, GCS 15. PEARL 4mm. | | |
| HEENT: | No facial edema | | |
| CVS: | Normal heart sounds. Pulses strong but rapid. Cap refill <2 seconds. | | |
| RESP: | Stridor at rest. Suprasternal retractions, poor air entry bilaterally. | | |
| ABDO: | Soft, non-tender, no rebound or peritonitis. | | |
| GU: | Nil | | |
| MSK: | Nil | SKIN: | Nil |



## Section VI: Scenario Progression

| Scenario States, Modifiers and Triggers | | | |
|---|---|---|---|
| Patient State | Patient Status | Learner Actions, Modifiers & Triggers to Move to Next State | |
| **1. Baseline State**<br>Rhythm: Sinus tach<br>HR: 180/min<br>BP: 80/46<br>RR: 65/min<br>$O_2$SAT: 90 % RA<br>T: 38.4°C | Significant work of breathing with stridor, suprasternal retractions, poor air entry bilaterally | <u>Learner Actions</u><br>-☐ Monitors, IV/IO access<br>-☐ Leave child in position of comfort with parent<br>-☐ Humidified O2 without agitating child (blow by)<br>-☐ Portable CXR<br>-☐ Antipyretic (Tylenol 180 mg)<br>-☐ Epi neb (5 mg/dose)<br>-☐ Dexamethasone (0.6 mg/kg) PO/IV/IM | <u>Modifiers</u><br>*Changes to patient condition based on learner action*<br>- O2 applied → SpO2 92%<br>- Neb epi→ HR 190, RR 60/min, still stridorous<br><br><u>Triggers</u><br>*For progression to next state*<br>- Epi nebs, humidified O2, dex, given → **2. Respiratory failure**<br>- 5 minutes→ **2. Respiratory failure** |
| **2. Respiratory failure**<br>HR: 150, weak peripheral pulses<br>BP: 75/35<br>RR: 12<br>$O_2$SAT: 80% | Drowsy, decreased respiratory rate, cyanotic, poor air entry bilaterally.<br>GCS 11 (E3 V3 M5) | <u>Learner Actions</u><br>-☐ Use BVM with 100% O2 + PEEP valve<br>-☐ Call Anesthesia, RT, ENT (ENT/Anesthesia unavailable)<br>-☐ Call for difficult airway cart, glide scope, needle cricothyroidotomy supplies<br>-☐ Multiple ETT sizes present<br>-☐ Insert IV and draw labs (VBG, CBC, lytes, Cr, Blood Cx)<br>-☐ IV fluid bolus 20 ml/kg<br>-☐ Sedation meds drawn up [Ketamine (1.5-2mg/kg) or Etomidate (0.3 mg/kg)]<br>-☐ Paralytic drawn up [Sux (1.5mg/kg), Roc (1.2 mg/kg)]<br>-☐ Push dose epi at bedside (1mcg/kg bolus q2-3 min)<br>-☐ Atropine at bedside (0.02mg/kg)<br>-☐ Prep the neck<br>-☐ Attempt intubation | <u>Modifiers</u><br>- BVM with 100% O2, PEEP → difficulty bagging, SpO2 to 82%<br>- 1st intubation attempt → unsuccessful, SpO2 to 75%<br>- No intubation considered 2 mins into state→ RN states *"I really think you need to intubate."*<br><br><u>Triggers</u><br>- Second intubation attempt, or use of LMA → **3. Failed Intubation**<br>- No intubation attempt by 3 min → **4. Cardiac Arrest** |



| | | | |
|---|---|---|---|
| **3. Failed Intubation**<br>HR: 110<br>BP: 65/p<br>RR: 6<br>O2 sat: 65% | | <u>Learner Actions</u><br>-☐ Analgesia/sedation (if not given yet)<br>-☐ Prep and drape<br>-☐ 1% lidocaine (conscious, semi conscious pt)<br>-☐ Locate cricothyroid membrane with non-dominant hand<br>-☐ 16-18g angiocath attached to sterile saline syringe, progressed through cricothyroid membrane directed 30-45 angle caudally, aspirating, once air aspirated, catheter advanced<br>-☐ 3 ml Luer-lock syringe with plunger removed connected to 7.5mm ETT connector **OR**<br>-☐ 3mm ETT connector attached directly to catheter **OR**<br>-☐ Transtracheal jet ventilator | <u>Modifiers</u><br>- If cric not attempted after 1 min →RN says "Do you want the cric kit"<br><br><u>Triggers</u><br>- Failure to perform needle cricothyrotomy within 5 min→ **4. Cardiac arrest**<br>- Successful needle cric → **5. Stabilization** |
| **4. Cardiac Arrest**<br>Rhythm: Asystole<br>HR: 0<br>BP: 0/0<br>RR: 0 | Pulseless and unresponsive | <u>Learner Actions</u><br>-☐ CPR with 15:2 ratio<br>-☐ BVM with 100% FiO2 while 2nd operator does needle cric<br>-☐ Epi 0.01mg/kg q3 mins<br>-☐ Perform needle cricothyrotomy as outlined above (state 3) | <u>Modifiers</u><br>- BVM difficult despite PEEP: RT *"This patient is too difficult to bag, could there be airway edema?"*<br>- Intubation attempts unsuccessful<br>- If no Cric being performed by resident →RN says "Do you want the crit kit"<br><br><u>Triggers</u><br>- Cric performed → **5. Stabilization** |
| **5. Stabilization**<br>Rhythm: Sinus tach<br>HR: 130<br>BP: 75/35<br>RR: Bagged rate<br>O2 sat: 94% | | <u>Learner Actions</u><br>-☐ Continues bagging wih 100% O2 attached to angiocath<br>-☐ Confirm placement (breath sounds, ETCO2, CXR)<br>-☐ Initiate sedation<br>-☐ Consult PICU<br>-☐ Ceftriaxone (100mg/kg) iv | <u>Modifiers</u><br><br><br><u>Triggers</u><br>-PICU consult→ **END CASE** |



## Section VII: Supporting Documents, Laboratory Results, & Multimedia

| Laboratory Results | | | | | | |
|---|---|---|---|---|---|---|
| Na: 135 | K: 4.8 | Cl: 97 | $HCO_3$: 22 | BUN: 5.0 | Cr: 55 | Glu: 5.0 |
| Ca: 2.1 | Mg: 1.1 | $PO_4$: 0.9 | Albumin: 40 | | | |
| | | | | | | |
| VBG | pH: 7.24 | $PCO_2$: 56 | $PO_2$: 50 | $HCO_3$: 22 | Lactate: 3.5 | |
| | | | | | | |
| WBC: 15.5 | Hg: 120 | Hct: 34 | Plt: 227 | | | |
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## Section VIII: Debriefing Guide

| General Debriefing Plan | | | |
|---|---|---|---|
| ☐ Individual | ☒ Group | ☐ With Video | ☒ Without Video |

| Objectives | |
|---|---|
| Educational Goal: | To expose learners to a difficult airway scenario in an infant with necessary percutaneous cricothyroidotomy |
| CRM Objectives: | 1) Effectively and calmly lead a team through a failed airway algorithm necessitating percutaneous cricothyroidotomy<br>2) Demonstrate high quality closed-loop communication throughout infant resuscitation |
| Medical Objectives: | 1) Verbalize and work through the differential diagnosis of stridor<br>2) Identify the need for intubation in a stridorous patient<br>3) Prepare options for airway management if intubation fails<br>4) Demonstrate the technique for needle cricothyroidotomy |

### Sample Questions for Debriefing

1) How do you feel the team communicated throughout this case?
2) What is in the differential diagnosis for stridor in an infant?
3) How did it feel to progress through the failed airway algorithm? At what point was it recognized by the team? Was it early enough?
4) What are the indications for percutaneous cricothyroidotomy? Contraindications? Why can you not do a surgical cricothyroidotomy in this patient?
5) What are some ways to connect the BVM to the 16-18g angiocath after percutaneous needle cricothyroidotomy?
6) How does a jet ventilator work? What are the typical pressures required for 0- 4 years old, 5-11 years old, 12 years old? (answer: <20mmHg, <30mmHg, 50mmHg)

| Key Moments |
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| Initiation of early treatments for the stridorous child |
| Recognition and preparation for a potential difficult airway in the pediatric patient |
| Performing percutaneous needle cricothyroidotomy |